

Applications of Queues

Q. Discuss the significance of Queues in computer science. Elaborate on various real-world and system-level applications where the Queue data structure is essential.

> **Definition to Remember**

> A **Queue** is a linear data structure that follows the First In First Out (FIFO) principle. It allows elements to be added at one end (enqueue) and removed from the other end (dequeue) in the order they were added.

1. System-Level Applications (Operating Systems & Networks)

Application	Description
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2. Real-World & Algorithmic Applications

Application	Description
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3. Print Spooler Example Diagram

```
graph LR
    User[User] --> Queue[Queue]
    Queue --> Printer[Printer]
```

[User :> Queue :> Printer]

> **Must-Write Points (for 10 marks)**

> 1. Q

> **Quick Recall**

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